

GOKUL KRISHNAN G

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PROFESSIONAL SUMMARY

AI-Native Software Engineer with **1 year of cumulative engineering experience** (including **8+ months in production AI development** at AVASOFT and prior cloud/full-stack internships), holding an academic foundation in **B.Tech in Artificial Intelligence & Data Science**. Core expertise spans **agent orchestration (LangGraph)**, **multi-modal LLM pipelines (AWS Bedrock, Claude Vision)**, **computer vision (OpenCV, TensorFlow)**, and **high-performance microservices (Python FastAPI, Go, PostgreSQL/pgvector)**. Experienced in taking distributed AI architectures from concept to production delivery.

TECHNICAL SKILLS

Languages: Python, Go (Golang), Java, JavaScript, HTML5, CSS3, SQL
AI & Agentic: LangGraph, LangChain, LlamaIndex, FastMCP (MCP), SerpAPI, RAG Pipelines, HITL, Agent Orchestration
ML & Computer Vision: TensorFlow, OpenCV, Image Processing, Model Training & Fine-Tuning, Scikit-Learn, SBERT
LLMs & AI APIs: AWS Bedrock (Claude Haiku 4.5 Vision, Titan, Nova), OpenAI APIs, Ollama (TinyLlama), Sentence-Transformers
Prompt Engineering: System prompts, few-shot design, chain-of-thought, structured JSON output, multi-turn context
Backend & APIs: FastAPI, Gorilla Mux, Node.js, Pydantic, RESTful APIs, SOLID Principles, Dependency Injection
Cloud & Infra: AWS (VPC, Public/Private Subnets, ECS, EventBridge, SES, S3, RDS, CloudWatch, EC2, IAM), Oracle OCI
Databases & Vector: PostgreSQL (pgvector), MySQL, SQLite, MS SQL Server, GORM, Asyncpg, AsyncPostgresSaver
CI/CD & DevOps: Azure DevOps (pipelines, PR webhooks), Docker, Kubernetes (master/worker arch.), GitHub Actions

PROFESSIONAL EXPERIENCE

AVASOFT

AI Analyst / Software Engineer (Full-Time — Joined Mid Final Year)

Dec 2025 – Aug 2026

Chennai, India

Project 1: AI-Powered Ticketing & Multi-Modal Automation Platform

Music & Live Sports Client

- Sole backend engineer on an event-driven microservice ecosystem (Python FastAPI & Go) with **50+ REST APIs** processing **30,000+ monthly emails**, integrated with client cloud databases and **Ticketmaster APIs**.
- Engineered a **multi-modal AI pipeline** using **Claude Haiku 4.5 Vision (AWS Bedrock)** and **OpenCV** to OCR-parse emails and attachments, extracting ticket counts, event dates, and discrepancies with **>97% accuracy**.
- Automated intent-based **Jira ticket routing** and email replies via **AWS EventBridge, SES, ECS** with Okta + JWT auth; slashed response times from **4+ hours to <45 seconds**.
- Integrated **Azure DevOps** CI/CD pipelines with **webhook-triggered automated builds and deployments** on PR events.

Project 2: T3 – AI Assessment & Developer Evaluation Platform

Internal Product

- Voice AI Interviews:** Agentic pipeline — speech-to-text transcription with LLM-based evaluation against question benchmarks.
- MCQ & Live Coding:** Adaptive LLM question generation & auto-scoring; real-time **AST syntax/logic validation** via FastAPI.

TECHNICAL PROJECTS — AVASOFT Production POCs

Autonomous Multi-Agent Code Generation & Review System | *LangGraph · AWS Bedrock · PostgreSQL · AST*

- Stateful multi-agent LangGraph system** — Orchestrator, Coder, Reviewer, Researcher agents with **AST static analysis** tools, cyclic self-healing loops, and **AsyncPostgresSaver** state persistence.
- End-to-end: problem intake → AI code generation → automated review → web research retrieval in one LangGraph graph.

RAG Knowledge Base & Computer Vision Pipeline | *TensorFlow · OpenCV · FastAPI · LlamaIndex · pgvector*

- Engineered a custom **TensorFlow & OpenCV image pre-processing engine** paired with a two-tier RAG API using **AWS Titan** embeddings and top-k cosine search over **PostgreSQL pgvector**.

Multi-Server MCP Coffee Shop Assistant | *LangChain · FastMCP · AWS Bedrock · PostgreSQL · Pydantic*

- Built an agentic assistant using **MultiServerMCPClient** to dynamically load tools, prompts, and resources from a remote **FastMCP** server over streamable HTTP.
- Integrated advanced LangChain middlewares: **SummarizationMiddleware** and **HumanInTheLoopMiddleware** with tool interrupts (`place_order`) and Postgres persistence via **AsyncPostgresSaver**.

INTERNSHIPS

Edu-Varsity <i>Cloud Computing Intern</i> Project: Cloud-Native Resume Enhancer Web Platform	Jun 2024 – Jul 2024 Chennai, India <i>Tech: AWS (VPC, ECS, RDS, CloudWatch, S3, IAM)</i>
- Architected and deployed a multi-tier Resume Enhancer platform on AWS ECS in a custom VPC with isolated public subnets (ingress/ALB) and private subnets for compute containers and Amazon RDS PostgreSQL. - Engineered secure resume document storage using Amazon S3 pre-signed URLs and configured Amazon CloudWatch metric alarms, log groups, and health checks for continuous container telemetry and automated recovery.	
FLASH INFOLABS <i>Full Stack Developer Intern</i> Project: Enterprise Employee Management & Directory Portal	Oct 2023 – Nov 2023 Chennai, India <i>Tech: Python FastAPI, JavaScript, HTML5, CSS3, REST APIs</i>
- Engineered a full-stack employee organization portal in a dedicated Git repo, designing modular backend REST APIs using Python FastAPI with Pydantic request models, relational schema queries, and role-based access validation. - Built a dynamic, responsive front-end interface using HTML5, CSS3, and modern JavaScript (Fetch API) for directory search, department categorization, and employee record lifecycle management.	

COLLEGE PROJECTS — github.com/GOKULKRISHNAN2044

AI-Powered Psychometric Assistant for Stress <i>Python · SBERT · DistilBERT · Gemini API · Rasa</i>	 GitHub
SBERT adaptive questioning + DistilBERT stress classification + Gemini API emotional tone analysis + Rasa chatbot with real-time speedometer visualization.	
Automated Guided Vehicle (AGV) — Intelligent Navigation <i>Python · Computer Vision · Path Planning</i>	 GitHub
Python CV + path-planning AGV system; open-source repository forked by 2 external contributors.	

ACHIEVEMENTS & RESEARCH PUBLICATIONS

 1st Prize — Ideathon RESONANCE '25, Rajalakshmi Engineering College (IEEE Computer Society Chapter)	Feb 2025
 2 International Research Publications IJFMR (IF: 9.24, 2025) & TIJER (IF: 8.57, Jul 2024)	2024 – 2025
 IEEEExtreme 18.0 Global programming competition, ranked among 19,000+ participants	Oct 2024

EDUCATION

Panimalar Engineering College <i>B.Tech in Artificial Intelligence & Data Science (CGPA: 8.33 / 10)</i>	2022 – 2026 Chennai, India
Jayam Metric Higher Secondary School <i>Higher Secondary Certificate — Computer Science (88.34%)</i>	May 2022 Dharmapuri, India

KEY CERTIFICATIONS

 Oracle Cloud Infrastructure 2024 Generative AI Certified Professional Oracle University	Jul 2024 – Jul 2026
 NPTEL — Introduction to Internet of Things IIT Kharagpur / SWAYAM (Elite Certificate)	2024
 Machine Learning Training AcmeGrade × IIT Bombay Mood Indigo	2023